

Design and Implementation of a Solar Panel Voltage Monitoring System Using Arduino

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Abstract—This project presents the design and implementation of a low-cost solar panel voltage monitoring system using an Arduino Uno microcontroller. A voltage divider circuit was used to safely measure the output voltage of a solar panel. Multiple voltage divider configurations were investigated and compared under indoor and outdoor lighting conditions. The project highlights the importance of calibration, voltage scaling, and experimental validation in renewable energy monitoring systems.

Index Terms—Arduino, Solar Energy, Renewable Energy, Voltage Divider, Embedded Systems, Monitoring System

I. INTRODUCTION

II. PROJECT OBJECTIVES

- Measure solar panel voltage using Arduino.
- Design a safe voltage divider circuit.
- Compare different voltage divider configurations.
- Investigate the influence of lighting conditions on solar panel voltage.
- Develop practical skills in renewable energy monitoring.

III. MATERIALS AND COMPONENTS

TABLE I
PROJECT COMPONENTS

Component	Quantity
Arduino Uno	1
Solar Panel	1
Breadboard	1
10k Ω Resistor	2
5.1k Ω Resistor	1
Jumper Wires	Several
USB Cable	1

IV. THEORY

A. Solar Panel Voltage

B. Arduino Analog Input

C. Voltage Divider Principle

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2} \quad (1)$$

V. SYSTEM DESIGN

A. Circuit Description

B. Experimental Setup

Figure 1 shows the complete outdoor setup used during testing.

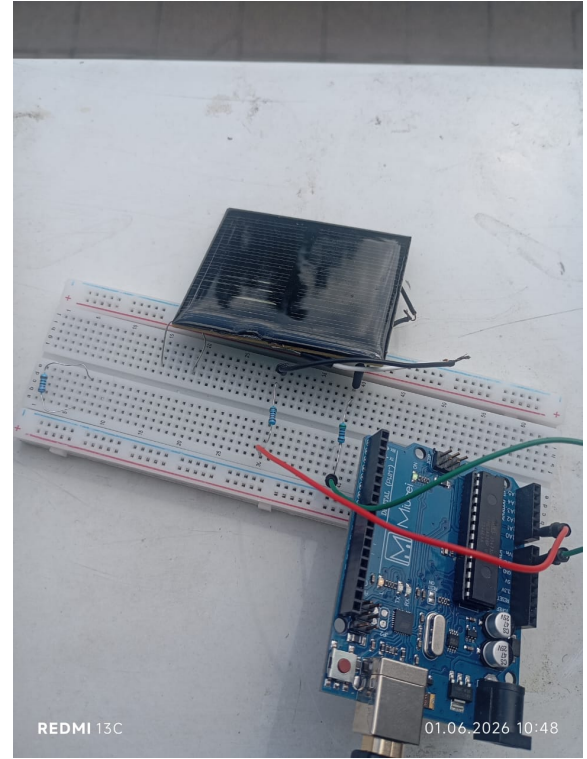


Fig. 1. Outdoor experimental setup showing the Arduino Uno, voltage divider circuit, and solar panel.

C. Development Stages

The project was completed through several iterations:

- 1) Initial design using a 10k Ω and 4.7k Ω voltage divider.
- 2) Substitution of the unavailable 4.7k Ω resistor with a 5.1k Ω resistor.
- 3) Discovery of a calibration mismatch between hardware and software.
- 4) Correction of the voltage scaling factor.
- 5) Evaluation of a safer 10k Ω and 10k Ω divider configuration.

6) Comparison of both voltage divider configurations.

VI. EXPERIMENTAL PROCEDURE

- 1) Assemble the voltage divider circuit.
- 2) Connect the solar panel to the divider.
- 3) Connect the divider output to Arduino analog pin A0.
- 4) Upload the Arduino measurement program.
- 5) Open the Serial Monitor.
- 6) Transfer measurement data to Excel.
- 7) Record indoor measurements.
- 8) Record outdoor measurements.
- 9) Compare the collected results.

VII. RESULTS

The voltage measurements were recorded using Microsoft Excel through serial communication with the Arduino.

A. Configuration A: $10k\Omega$ and $5.1k\Omega$

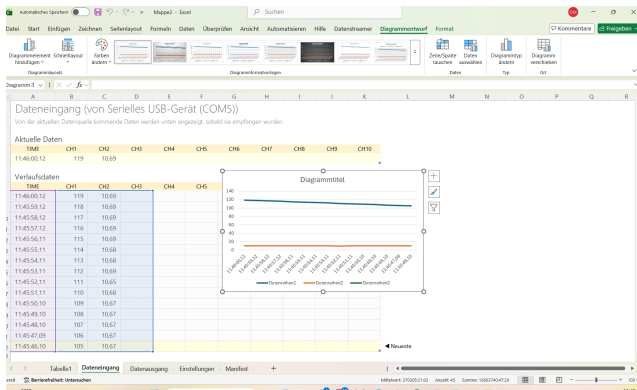


Fig. 2. Indoor measurements using the $10k\Omega$ and $5.1k\Omega$ configuration.

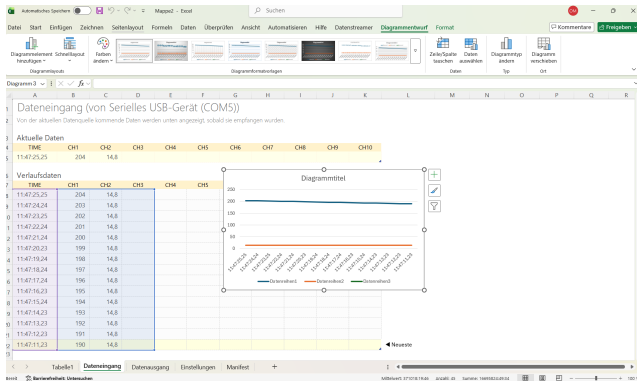


Fig. 3. Outdoor measurements using the $10k\Omega$ and $5.1k\Omega$ configuration.

TABLE II
CONFIGURATION A RESULTS

Condition	Voltage (V)
Indoor	10.69
Outdoor	14.80

B. Configuration B: $10k\Omega$ and $10k\Omega$

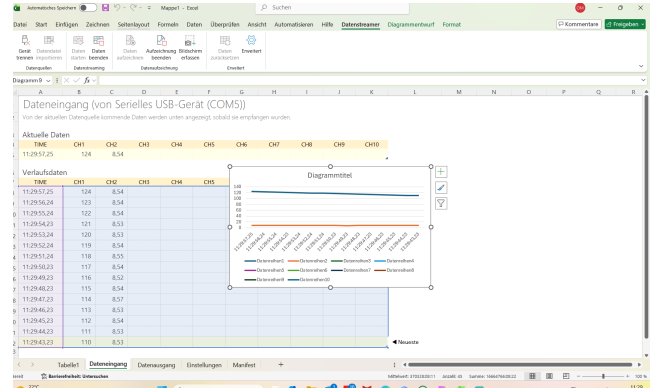


Fig. 4. Indoor measurements using the $10k\Omega$ and $10k\Omega$ configuration.

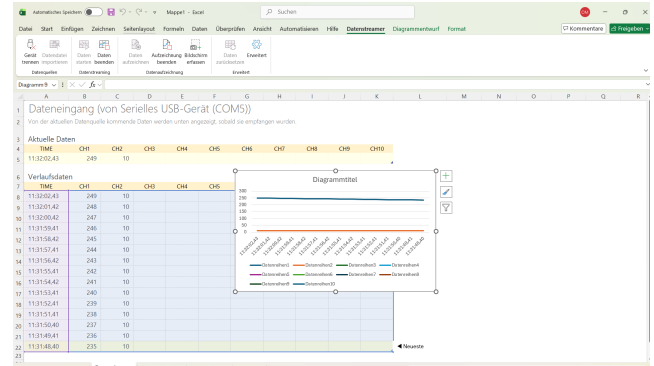


Fig. 5. Outdoor measurements using the $10k\Omega$ and $10k\Omega$ configuration.

TABLE III
CONFIGURATION B RESULTS

Condition	Voltage (V)
Indoor	8.54
Outdoor	10.00

C. Comparison of Configurations

TABLE IV
COMPARISON OF VOLTAGE DIVIDER CONFIGURATIONS

Configuration	Indoor (V)	Outdoor (V)
$10k\Omega + 5.1k\Omega$	10.69	14.80
$10k\Omega + 10k\Omega$	8.54	10.00

VIII. DISCUSSION

IX. CHALLENGES AND LESSONS LEARNED

- A $5.1k\Omega$ resistor was used instead of the planned $4.7k\Omega$ resistor.
- The software initially used the wrong scaling factor.
- Calibration errors led to inaccurate measurements.
- Updating the software to match the hardware improved accuracy.

- Different voltage divider configurations affect safety and sensitivity.
- Hardware and software must always be consistent.

X. FUTURE WORK

- LCD/OLED display integration.
- Battery voltage monitoring.
- Solar battery charge controller.
- Current measurement and power calculation.
- Data logging to SD card.
- Wireless monitoring using Wi-Fi.
- Smart solar battery charge monitor.

XI. CONCLUSION

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